

Before beginning this book, readers should be familiar with the contents of the *Junior Secondary School* course and Book 1 of the *Senior Secondary School* course. References to earlier books in the series are given at the end of each section.

Number and numeration

- a Numbers are usually written in the decimal **place value** system (Fig. P1).

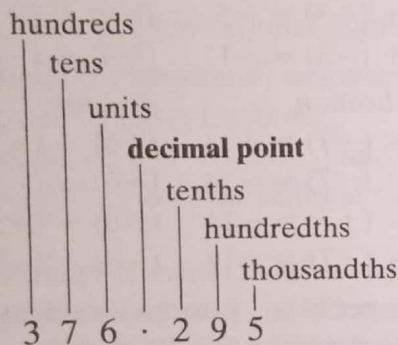


Fig. P1

The symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 are called **digits**.

- b $28 \div 7 = 4$. 7 is a **whole number** which divides exactly into another whole number, 28. 7 is a **factor** of 28. 28 is a **multiple** of 7.
- c A **prime number** has only two factors, itself and 1. 1 is *not* a prime number. 2, 3, 5, 7, 11, 13, 17, ..., are prime numbers. They continue without end. The **prime factors** of a number are those factors which are prime. For example, 2 and 5 are the prime factors of 40. 40 can be written as a **product of prime factors**; either $2 \times 2 \times 2 \times 5 = 40$, or in **index form**, $2^3 \times 5 = 40$.
- d The numbers 18, 24 and 30 all have 3 as a factor. 3 is a **common factor** of the numbers. The **highest common factor** (HCF) is the largest of the common factors of a given set of numbers. For example, 2, 3 and 6 are the

common factors of 18, 24 and 30; 6 is the HCF.

The number 48 is a multiple of 4 and a multiple of 6. 48 is a **common multiple** of 4 and 6. The **lowest common multiple** (LCM) is the smallest of the common multiples of a given set of numbers. For example, 12 is the LCM of 4 and 6.

- e A **fraction** is the number obtained when one number (the **numerator**) is divided by another number (the **denominator**). The fraction $\frac{5}{8}$ means $5 \div 8$ (Fig. P2).

$$\begin{array}{l} \text{numerator} \\ 5 \\ \hline \text{dividing line} \\ 8 \\ \text{denominator} \end{array}$$

Fig. P2

Fractions are used to describe parts of quantities (Fig. P3).

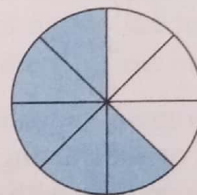


Fig. P3 $\frac{5}{8}$ of the circle is shaded

The fractions $\frac{5}{8}$, $\frac{10}{16}$, $\frac{15}{24}$ all represent the same amount; they are **equivalent fractions**. $\frac{5}{8}$ is the **simplest form** of $\frac{15}{24}$, i.e. $\frac{15}{24}$ in its **lowest terms** is $\frac{5}{8}$.

To add or subtract fractions, change them to equivalent fractions with a **common denominator**. For example,

$$\begin{aligned} \frac{5}{8} + \frac{2}{3} &= \frac{15}{24} + \frac{16}{24} = \frac{15 + 16}{24} = \frac{31}{24} \quad (= 1\frac{7}{24}) \\ \frac{13}{16} - \frac{5}{8} &= \frac{13}{16} - \frac{10}{16} = \frac{13 - 10}{16} = \frac{3}{16} \end{aligned}$$

To multiply fractions, multiply numerator by numerator and denominator by denominator. For example,

$$\frac{5}{8} \times \frac{2}{3} = \frac{5 \times 2}{8 \times 3} = \frac{10}{24} \quad \left(= \frac{5}{12} \text{ in simplest form} \right)$$

$$12 \times \frac{5}{8} = \frac{12}{1} \times \frac{5}{8} = \frac{12 \times 5}{1 \times 8} = \frac{60}{8} \left(= \frac{15}{2} = 7\frac{1}{2} \right)$$

To divide by a fraction, multiply by the **reciprocal** of the fraction. For example,

$$35 \div \frac{5}{8} = \frac{35}{1} \times \frac{8}{5} = \frac{35 \times 8}{1 \times 5} = \frac{7 \times 8}{1} = 56$$

$$\frac{5}{8} \div 3\frac{3}{4} = \frac{5}{8} \div \frac{15}{4} = \frac{5}{8} \times \frac{4}{15} = \frac{5 \times 4}{8 \times 15} = \frac{20}{120} \quad \left(= \frac{1}{6} \right)$$

- f $x\%$ is short for $\frac{x}{100}$. 64% means $\frac{64}{100}$. To change a fraction to an equivalent percentage, multiply the fraction by 100. For example, $\frac{5}{8}$ as a percentage $= \frac{5}{8} \times 100\% = \frac{125}{2}\% = 62\frac{1}{2}\%$.

- g To change a fraction to a **decimal fraction**, divide the numerator by the denominator. For example,

$$\frac{5}{8} = 0.625$$

$$\begin{array}{r} 0.625 \\ 8 \overline{) 5.000} \\ \underline{48} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 0 \end{array}$$

When adding or subtracting decimals, write the numbers in a column with the decimal points exactly under each other. For example: Add 2.29, 0.084 and 4.3, then subtract the result from 11.06.

$$\begin{array}{r} 2.29 \\ 0.084 \\ + 4.3 \\ \hline 6.674 \end{array} \quad \begin{array}{r} 11.06 \\ - 6.674 \\ \hline 4.386 \end{array}$$

To multiply decimals, first ignore the decimal points and multiply the given numbers as if they are whole numbers. Then place the decimal point so that the answer has as many digits after the point as there are in the given numbers together. For example, 0.08×0.3 .

$$8 \times 3 = 24$$

There are three digits after the decimal points in the given numbers, so, $0.08 \times 0.3 = 0.024$.

To divide by decimals, make an equivalent division such that the divisor is a whole number. For example, $5.6 \div 0.07$.

$$5.6 \div 0.07 = \frac{5.6}{0.07} = \frac{5.6 \times 100}{0.07 \times 100} = \frac{560}{7} = 80$$

- h Numbers may be positive or negative. Positive and negative numbers are called **directed numbers**. Directed numbers can be shown on a **number line** (Fig. P4).

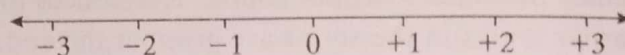


Fig. P4

The following examples show how to add, subtract, multiply and divide directed numbers.

<i>addition</i>	<i>subtraction</i>
$(+8) + (+3) = +11$	$(+9) - (+4) = +5$
$(+8) + (-3) = +5$	$(+9) - (-4) = +13$
$(-8) + (+3) = -5$	$(-9) - (+4) = -13$
$(-8) + (-3) = -11$	$(-9) - (-4) = -5$
<i>multiplication</i>	<i>division</i>
$(+2) \times (+7) = +14$	$(+6) \div (+3) = +2$
$(+2) \times (-7) = -14$	$(+6) \div (-3) = -2$
$(-2) \times (+7) = -14$	$(-6) \div (+3) = -2$
$(-2) \times (-7) = +14$	$(-6) \div (-3) = +2$

An **integer** is any positive or negative *whole* number as shown in Fig. P4 above.

- i The number 3.7×10^4 is in **standard form**. The first part of the product is between 1 and 10, and the second is a power of 10.
- j When **rounding off** numbers, the digits 1, 2, 3, 4 are rounded down, and the digits 5, 6, 7, 8, 9 are rounded up. For example,

$$\begin{aligned} 3425 &= 3430 \text{ to 3 significant figures (s.f.)} \\ &= 3400 \text{ to the nearest hundred} \\ 7.283 &= 7.28 \text{ to 2 decimal places} \\ &= 7.3 \text{ to 1 decimal place} \\ &= 7 \text{ to the nearest whole number} \end{aligned}$$

When calculating from 4-figure tables it is usual to round numbers to 3 s.f.

- k **Binary numbers** use a **base two** place value system. For example,

$$\begin{aligned} 110101_{\text{two}} &= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + \\ &\quad 1 \times 2^2 + 0 \times 2^1 + 1 \\ &= 32 + 16 + 0 + 4 + 0 + 1 = 53_{\text{ten}} \end{aligned}$$

To change a base ten number to base two, divide repeatedly by 2 until further division is impossible. The remainders in each line of working, reading upwards, give the required number. For example, Change 29_{ten} to a binary number.

		rem.
2	29	
2	14	1
2	7	0
2	3	1
2	1	1
	0	1

$29_{ten} = 11\ 101_{two}$

- 1 A **rational number** is any number in the form $\frac{a}{b}$, where a and b are whole numbers. For example, $\frac{1}{9}$, $\frac{15}{1}$, $\frac{33}{8}$, $\frac{5}{5}$ are rational numbers.

Non-rational (or irrational) numbers cannot be expressed as exact fractions.

For example, $\sqrt{2}$ ($=1.414\ 213\dots$) and π ($=3.141\ 592\dots$) are non-rational numbers.

Detailed coverage of *number* and *numeration* is given in the JSS course books as follows: Book 1, Chapters 1, 2, 3, 4, 9, 12, 23, 24; Book 2, Chapters 1, 2, 4, 8, 9, 10, 15, 18; Book 3, Chapters 1, 5, 11, 20, R1, and the SSS course Book 1, Chapters 1, 3, 13, 19, 20.

Review test 1 (Number and numeration)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 2.

- ① Express each of the following as a product of factors. Hence find their square roots.

a 19 600 b 625
c 1 024 d 5 776

- ② Express the following
i in standard form,
ii as a logarithm.

a 9 000 000 b 70 000
c 80 d 2

- ③ Round off the following to
i 1 significant figure,
ii 2 significant figures,
iii 3 significant figures.

a 7 607 b 42 824
c 31 499 d 0.007 407

- ④ Round off the following to
i 1 decimal place,
ii 2 decimal places,
iii 3 decimal places.

a 23.701 6 b 0.009 4
c 8.853 5 d 0.007 7

- ⑤ Round off the following to the nearest
i whole number,
ii tenth,
iii hundredth.

a 5.049 b 59.509

- ⑥ Simplify the following.

a $(-9) \times (-2)$ b $33 \times (-3)$
c $-51 \div 17$ d $-35 \div (-5)$

- ⑦ A school has 603 students. The average mass of a student is approximately 49 kg. Estimate to 1 significant figure, the total mass in tonnes of all the students in the school.

- ⑧ Divide the following quantities in the given ratios.

a ₦3 604 in the ratio 1 : 3
b 750 mm in the ratio 3 : 2
c 176 bananas in the ratio 2 : 4 : 5.

- ⑨ Find the discount price if

a 10% is given on a cost price of ₦600
b 25% is given on a cost price of ₦6 600.

- ⑩ Use tables to find:

a 170^2 b $2\ 900^2$ c $\sqrt{5\ 446}$ d $\sqrt{804}$

- ⑪ a Convert to base ten:

i $10\ 101_{two}$ ii 11.011_{two}

b Convert to base two:

i 23_{ten} ii $5\frac{3}{4}_{ten}$

- ⑫ Reduce the following to their simplest form.

a 44 (mod 5)
b $41 \pmod{3} \ominus 16 \pmod{3}$

Review test 2 (Number and numeration)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

- ① Express each of the following as a product of factors. Hence find their square roots.
a 8100 b 12544 c 1600 d 1936
- ② Express the following
i in standard form,
ii as a logarithm.
a 8000 b 400 000 000
c 78 d 5.1
- ③ Round off the following to
i 1 significant figure,
ii 2 significant figures,
iii 3 significant figures.
a 657.63 b 9.0175
c 69.487 d 0.08855
- ④ Round off the following to
i 1 decimal place,
ii 2 decimal places,
iii 3 decimal places.
a 46.3399 b 5.0275
c 0.0493 d 0.0084
- ⑤ Round off the following to the nearest
i whole number,
ii tenth,
iii hundredth.
a 0.625 b 8.385
- ⑥ Simplify the following.
a $-61 \times (-8)$ b -8×10
c $68 \div (-4)$ d $-72 \div (-8)$
- ⑦ A typist can type about 32 words per minute. Estimate, to 1 significant figure, how long it will take her to type a text that is 891 words long.
- ⑧ Divide the following quantities in the given ratios.
a ₦400 in the ratio 3:2
b 330 ml in the ratio 4:7
c 90 eggs in the ratio 3:3:4

- ⑨ Find the discount price if
a 5% is given on a cost price of ₦4 000
b 20% is given on a cost price of ₦680.
- ⑩ Use tables to find:
a 47^2 b 2080^2 c $\sqrt{4003}$ d $\sqrt{555}$
- ⑪ a Convert to base ten:
i $111\,001_{two}$ ii 10.111_{two}
b Convert to base two:
i 33_{ten} ii $2\frac{1}{2}_{ten}$
- ⑫ Reduce the following to their simplest terms.
a $39 \pmod{4}$
b $22 \pmod{5} \oplus 13 \pmod{5}$

Algebraic processes

- a $3y^2 + 2x - 7x$ is an example of an **algebraic expression**. The letters y and x stand for numbers. $3y^2$, $2x$ and $7x$ are the **terms** of the expression. $3y^2$ is short for $3 \times y \times y$. 3 is the **coefficient** of y^2 . Algebraic terms may be **simplified** by combining **like terms**. Thus, $3y^2 + 2x - 7x = 3y^2 - 5x$ since $2x$ and $7x$ are like terms (i.e. both terms in x).
- b $3(5x - 2) = 11x$ is an **algebraic sentence** containing an equals sign; it is an **equation** in x . x is the **unknown** of the equation. To **solve an equation** means to find the value of the unknown which makes the equation true. Use the **balance method** to solve equations. For example,

$$3(5x - 2) = 11x$$

Clear brackets.

$$15x - 6 = 11x$$

Subtract $11x$ from both sides.

$$15x - 11x - 6 = 11x - 11x$$

$$4x - 6 = 0$$

Add 6 to both sides.

$$4x - 6 + 6 = 0 + 6$$

$$4x = 6$$

Divide both sides by 4.

$$\frac{4x}{4} = \frac{6}{4}$$

$$x = 1\frac{1}{2}$$

In general, when solving equations, **i** first clear brackets and fractions, **ii** using equal additions and/or subtractions, collect unknown terms on one side of the equals sign and known terms on the other, **iii** where necessary, divide or multiply both sides of the equation by the same number to find the unknown.

c An **inequality** is an algebraic sentence which contains an inequality sign:

- < is less than
- ≤ is less than or equal to
- > is greater than
- ≥ is greater than or equal to
- ≠ is not equal to

Inequalities are solved in much the same way as equations. However, when both sides of an inequality are multiplied or divided by a negative number, the inequality sign is *reversed*. For example,

$$\text{If } -3a \leq 12$$

divide both sides by -3 and reverse the inequality,
then $a \geq -4$

d $2x - 5y = 16$ is a **linear equation** with two **variables**, x and y . There are many pairs of values of x and y which **satisfy** this equation (i.e. make the equation true). For example, if $x = 13$ and $y = 2$ or if $x = 8$ and $y = 0$, the equation will be true. Given two equations, such as $2x - 5y = 16$ and $x + 4y = -5$, it is usually possible to find values of x and y which satisfy *both* equations simultaneously (i.e. at the same time). To solve a pair of **simultaneous linear equations**, either use a graphical method or the method of elimination and substitution. For example,

$$2x - 5y = 16 \quad (1)$$

$$x + 4y = -5 \quad (2)$$

Multiply (2) by 2.

$$2x + 8y = -10 \quad (3)$$

Subtract (3) from (1) to eliminate terms in x .

$$-13y = 26$$

$$y = -2$$

Substitute -2 for y in (2) to find x .

$$x + 4(-2) = -5.$$

$$x - 8 = -5$$

$$x = 3$$

The solution is $x = 3$ and $y = -2$.

e A **graph** of an algebraic sentence is a picture representing the meaning of the sentence. Graphs of equations and inequalities in one variable can be shown on the number line (Fig. P5).

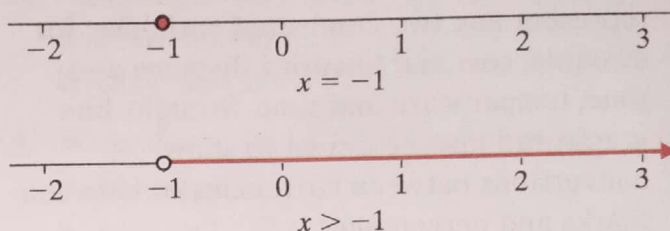


Fig. P5

For graphs connecting two variables, two **axes** are drawn at right angles to each other to give a **cartesian plane**. The horizontal **x-axis** and the vertical **y-axis** cross at their zero-point, the **origin** of the plane. Fig. P6 is the graph of the equation $y = 2x - 3$.

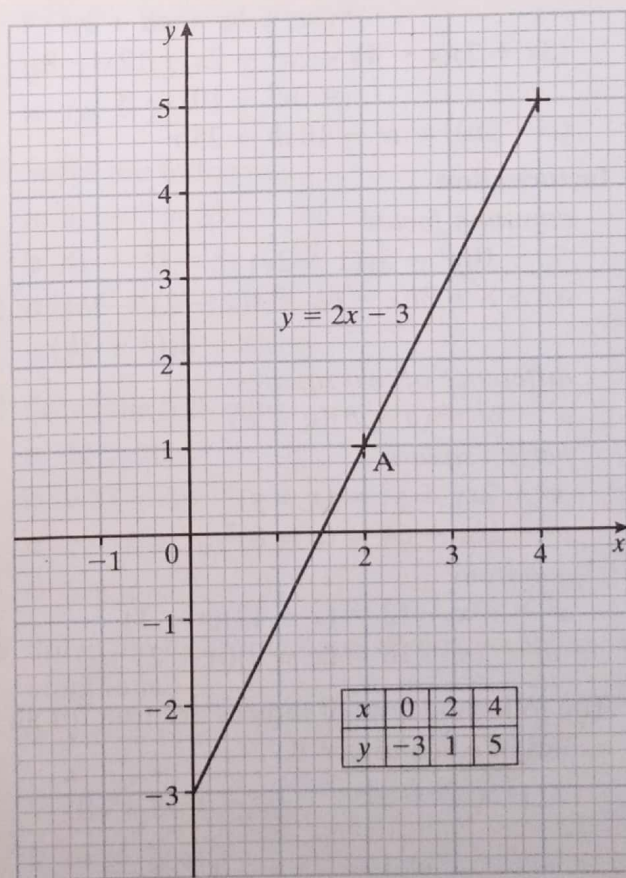


Fig. P6

To draw a straight-line graph, plot at least three points which satisfy the given equation. See the table of values in Fig. P6. At point A in Fig. P6, $x = 2$ and $y = 1$. The **coordinates** of A are A(2, 1). The **order** of the coordinates is important: the **x-coordinate** is given first, the **y-coordinate** second.

Straight-line graphs can be drawn to represent any two connected variables: for example, cost and quantity, distance and time, temperature and time. Straight-line graphs can also be drawn to show conversions between currencies or between marks and percentages.

- f Algebraic expressions may be **factorised** or **expanded** in accordance with the basic rules of arithmetic. Some examples are given below.

expansion

$$3(a - 2b) = 3a - 6b$$

$$(5 + 8x)x = 5x + 8x^2$$

$$(a + b)(c + d) = c(a + b) + d(a + b) \\ = ac + bc + ad + bd$$

$$(3x + 2)(x - 4) = 3x^2 + 2x - 12x - 8 \\ = 3x^2 - 10x - 8$$

$$(a - 5b)^2 = a^2 - 10ab + 25b^2$$

factorisation

common factor

$$5y - 10y^2 = 5y(1 - 2y)$$

$$4x - 8 + 3bx - 6b = 4(x - 2) + 3b(x - 2) \\ = (x - 2)(4 + 3b)$$

difference of two squares

$$a^2 - b^2 = (a + b)(a - b)$$

perfect square

$$x^2 + 2xy + y^2 = (x + y)^2$$

$$a^2 - 16a + 64 = (a - 8)^2$$

quadratic

$$x^2 + 13x + 12 = (x + 12)(x + 1)$$

$$n^2 - 7n - 18 = (n - 9)(n + 2)$$

- g The following laws of indices are true for all values of a , b and $x \neq 0$.

$$x^a \times x^b = x^{a+b} \quad x^a \div x^b = x^{a-b}$$

$$x^0 = 1 \quad x^{-a} = \frac{1}{x^a}$$

- h If x is **directly proportional** to y , then $x \propto y$ or $x = ky$ (where k is a constant). This is an example of **direct variation**.

If x is **inversely proportional** to y , then $x \propto \frac{1}{y}$ or $x = \frac{K}{y}$ (where K is a constant). This is an example of **inverse variation**.

If x is directly proportional to y and inversely proportional to z , then $x \propto \frac{y}{z}$ or

$x = \frac{Cy}{z}$ (where C is a constant). This is an example of **joint variation**.

Detailed coverage of *algebraic processes* is given in the JSS course books as follows: Book 1, Chapters 5, 7, 10, 15, 19; Book 2, Chapters 5, 7, 11, 12, 13, 14, 22, 23; and Book 3, Chapters 2, 3, 6, 9, 10, 13, 14, 19, R2, and the SSS course Book 1, Chapters 4, 5, 6, 7, 8, 10, 18.

Review test 3 (Algebraic processes)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 4.

- ① State the additive inverse of the following.
a -6 b $+23$ c -227 d $+\frac{4}{11}$
- ② State the multiplicative inverse of the following.
a $+6$ b $-\frac{1}{8}$ c $+48$ d -0.5
- ③ Remove the brackets and simplify.
a $3(2a - b) - 5(2a - 4b)$
b $y(5y - 3) - 3(y - 4)$
- ④ Expand the following.
a $(5 - x)(7 + x)$ b $(3x - y)(2x - 5y)$
- ⑤ Find the HCF of the following.
a $12x^2$ and $30x^2$ b $24xy^2$ and $6x^2y$
- ⑥ Write down the LCM of the following.
a pq and qr b $2a^2$ and $7ab$

⑦ Factorise the following.

- a $2a^2 - 9ab$
- b $-18pq - 12p$
- c $9x + 6 - 3x^2 - 2x$
- d $3a - cb - 3b + ac$
- e $x^2 - 8x - 20$
- f $100 - 49x^2$
- g $4x^2 - 11x - 20$
- h $3x^2 + 7x - 20$

⑧ Solve the following.

- a $53 = 5 + 6x$
- b $6x - 1 = 24 + x$
- c $x^2 - 11x + 24 = 0$
- d $2x^2 + 9x - 5 = 0$

⑨ Solve the following.

- a $2x < 14$
- b $-20 > 5x$
- c $5x - 12 \leq 7x$
- d $6 - 8x > 41 - 3x$

⑩ Solve the following pairs of simultaneous equations.

- a $y = 2x - 4$
 $3x + y = 11$
(by substitution)
- b $5x - 2y = 14$
 $2x + 2y = 14$
(by elimination)

⑪ Make h the subject of the equation $V = \pi r^2 h$. Hence find h when $V = 450 \text{ cm}^3$ and $r = 5 \text{ cm}$. Use the value 3 for π .

⑫ If $\mathcal{E} = \{1, 2, 3, 4, \dots, 10\}$, $P = \{\text{odd numbers}\}$ and $Q = \{\text{multiples of 3}\}$, list the members of

- a $P \cup Q$
- b $P' \cap Q'$
- c $\mathcal{E} \cap Q$

Review test 4 (Algebraic processes)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher.

① State the additive inverse of

- a $+2.5$
- b -97
- c $+446$
- d -0.73

② State the multiplicative inverse of

- a 3
- b $\frac{9}{10}$
- c -14
- d -0.4

③ Remove the brackets and simplify.

- a $7x(2x - 1) - 5(3x - 9)$
- b $a(2a + b) - 2b(a + 5b)$

④ Expand the following.

- a $(x + 7)(x - 3)$
- b $(3x - 7)(4x + 1)$

⑤ Find the HCF of the following.

- a $24xy$ and $8xy$
- b $18xy^2$ and $12x^2y$

⑥ Write down the LCM of the following.

- a $3a$ and $8b$
- b $15xy$ and $2x^2y^2$

⑦ Factorise the following.

- a $5a + 5z$
- b $2a^2 + 10a$
- c $8m - 2 + 4nm - n$
- d $6a + bm + 6b + am$
- e $x^2 - 5x + 6$
- f $4d^2 - 9e^2$
- g $6x^2 - 13x + 6$
- h $7x^2 + 25x - 12$

⑧ Solve the following.

- a $8g - 7 = 33$
- b $4x - 20 = 5x - 22$
- c $x^2 - 2x - 15 = 0$
- d $3x^2 - 10x + 7 = 0$

⑨ Solve the following.

- a $4x > 20$
- b $-13 < 3x$
- c $x + 2 \geq 10x - 25$
- d $5 - x < 17 - 3x$

⑩ Solve the following pairs of simultaneous equations.

- a $y = x + 1$
 $x + y = 3$
(by substitution)
- b $5x + 3y = 1$
 $2x + 3y = -5$
(by elimination)

- ⑪ I varies directly with E and inversely with R .
If

$I = 3$ when $E = 240$ and $R = 40$, find the equation that connects I , E and R . Hence find I when $E = 120$ and $R = 30$.

- ⑫ If $\mathbb{E} = \{\text{positive integers}\}$, list the members of the following sets.

- a $\{x : x > 9\}$
- b $\{x : x > -9\}$
- c $\{x : 2x = 11\}$

Geometry and mensuration

- a Fig. P7 gives sketches and names of some common solids.

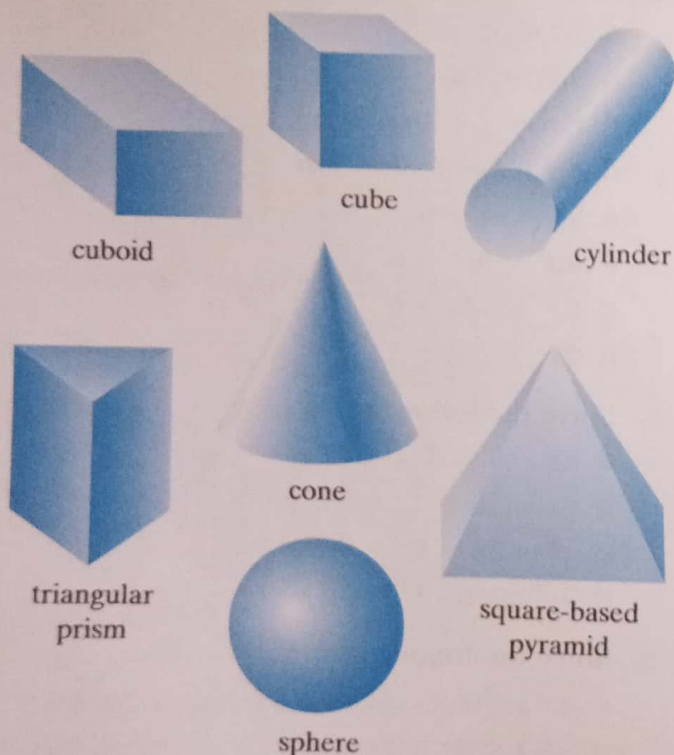


Fig. P7

All solids have **faces**; most solids have **edges** and **vertices** (Fig. P8).

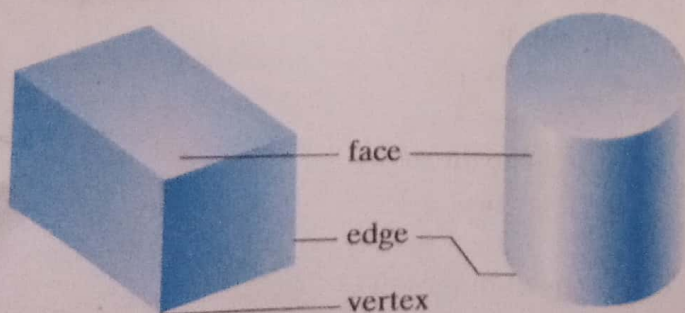


Fig. P8

Formulae for the **surface area** and **volume** of common solids are given in Table P1.

	surface area	volume
cube edge s	$6s^2$	s^3
cuboid length l , breadth b , height h	$2(lb + bh + lh)$	lbh
prism height h , base area A		Ah
sphere radius r	$4\pi r^2$	$\frac{4}{3}\pi r^3$
cylinder base radius r , height h	$2\pi r(r + h)$	$\pi r^2 h$
cone base radius r , height h , slant height l	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$

Table P1

Fig. P9 shows two common methods of **drawing** a cuboid. These methods may be used with any solid.

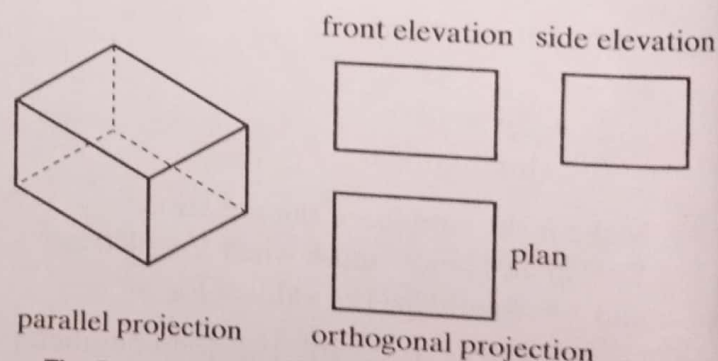


Fig. P9

In **parallel projections**, lines that are parallel on the solid appear parallel on the drawing. In **orthogonal projections**, the solid is represented by separate scale drawings of its **plan** and **elevations**.

The **net** of a solid is the plane shape which can be folded to make the solid. The net of a cube is shown in Fig. P10.

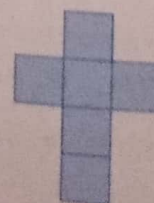


Fig. P10

b Angle is a measure of rotation or turning.

1 **revolution** = 360 **degrees** (1 rev = 360°)

1 **degree** = 60 **minutes** ($1^\circ = 60'$)

The names of angles change with their size.
See Fig. P11.

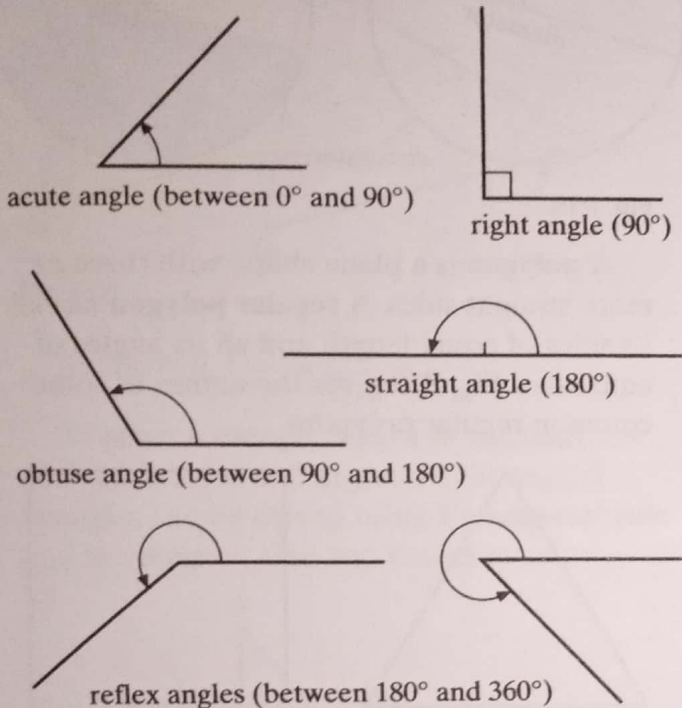


Fig. P11

Angles are measured and constructed using a **protractor**.

Fig. P12 shows some properties of angles formed when straight lines meet.

e) Corresponding angles are equal. $a = b$ and $p = q$

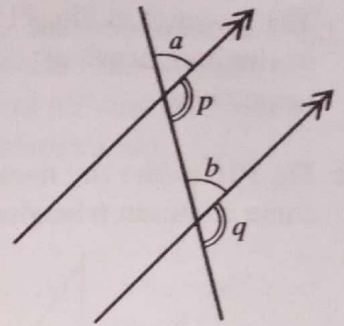


Fig. P12

In Fig. P13, α is the **angle of elevation** of the top of the flag-pole from the girl and β is the **angle of depression** of the girl from the boy.

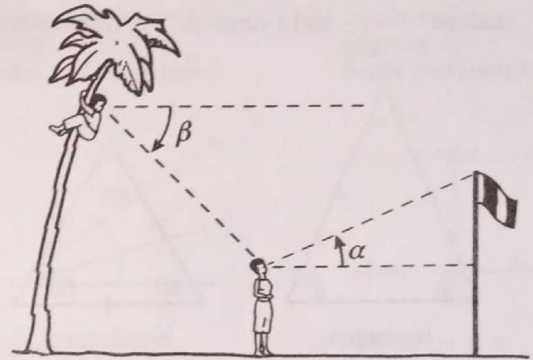


Fig. P13

Directions are taken from the **points of the compass** (Fig. P14).

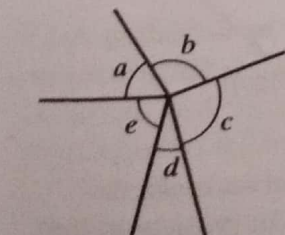


Fig. P14

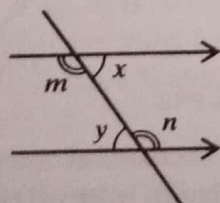
A **3-figure bearing** is a direction given as the number of degrees from north measured in a clockwise direction. See Fig. P15.

a) The sum of the angles on a straight line is 180° .
 $a + b + c = 180^\circ$

b) Vertically opposite angles are equal.
 $p = q$ and $r = s$



c) The sum of the angles at a point is 360° .
 $a + b + c + d + e = 360^\circ$



d) Alternate angles are equal. $x = y$ and $m = n$

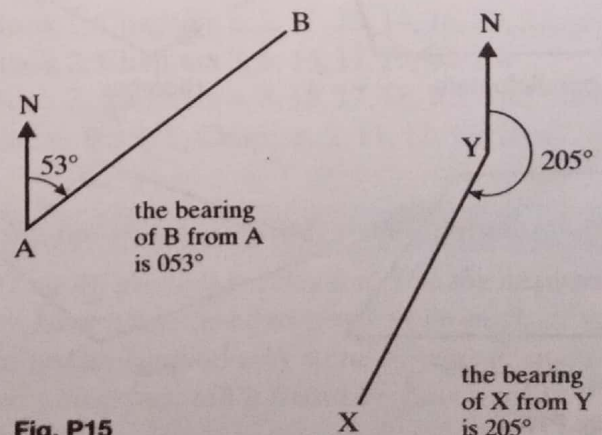


Fig. P15

The bearings in Fig. P15 can also be given as **compass bearings**: N53°E and S25°W respectively.

- c Fig. P16 shows the names and properties of some common **triangles**.

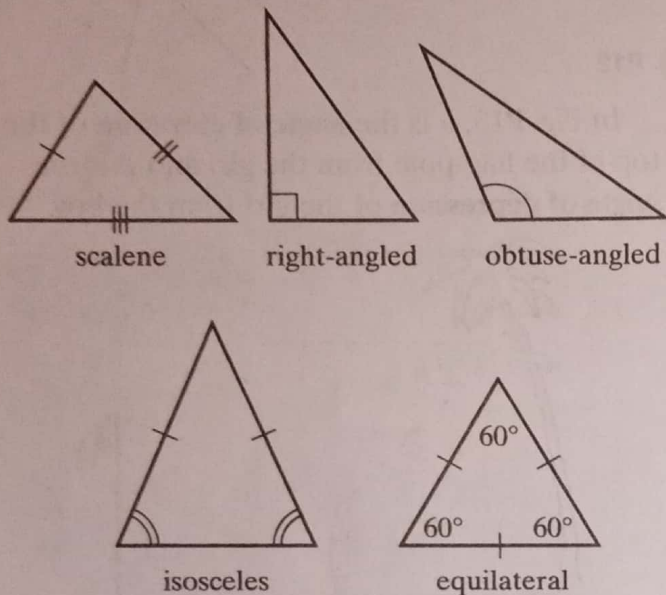


Fig. P16 Some common triangles

Fig. P17 shows the names and properties of some common **quadrilaterals**.

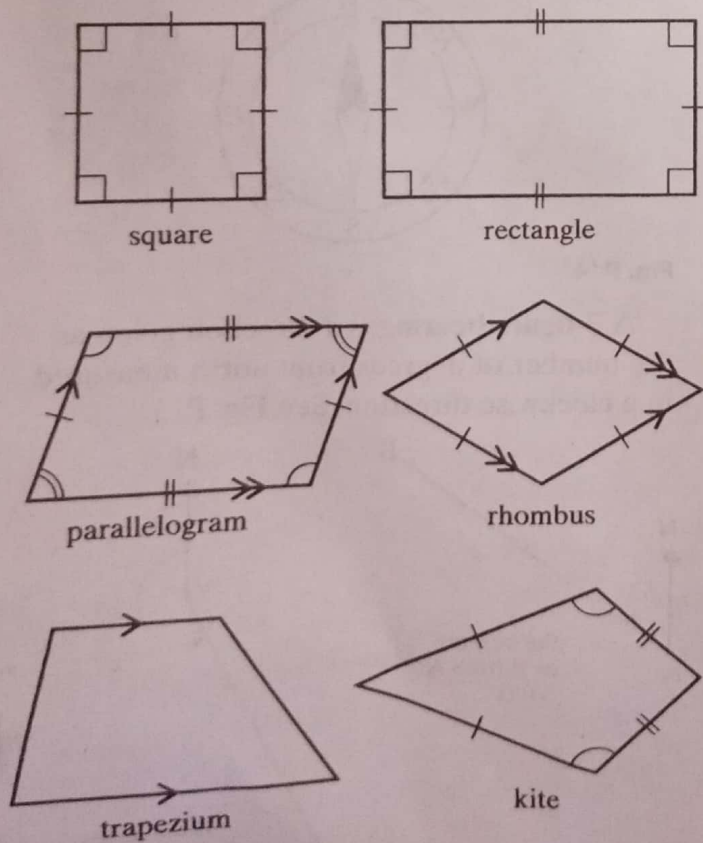


Fig. P17

Fig. P18 gives the names of lines and regions in a **circle**.

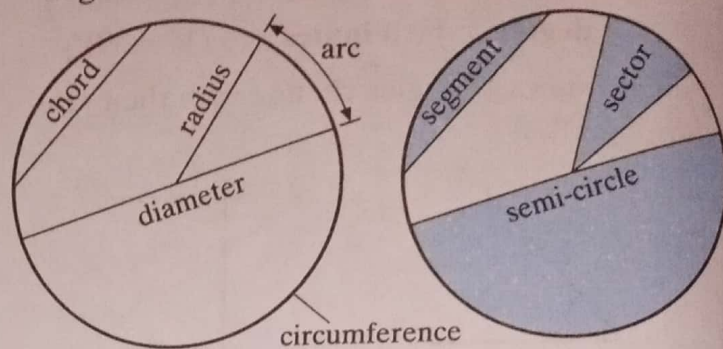


Fig. P18

A **polygon** is a plane shape with three or more straight sides. A **regular polygon** has all its sides of equal length and all its angles of equal size. Fig. P19 gives the names of some common regular polygons.

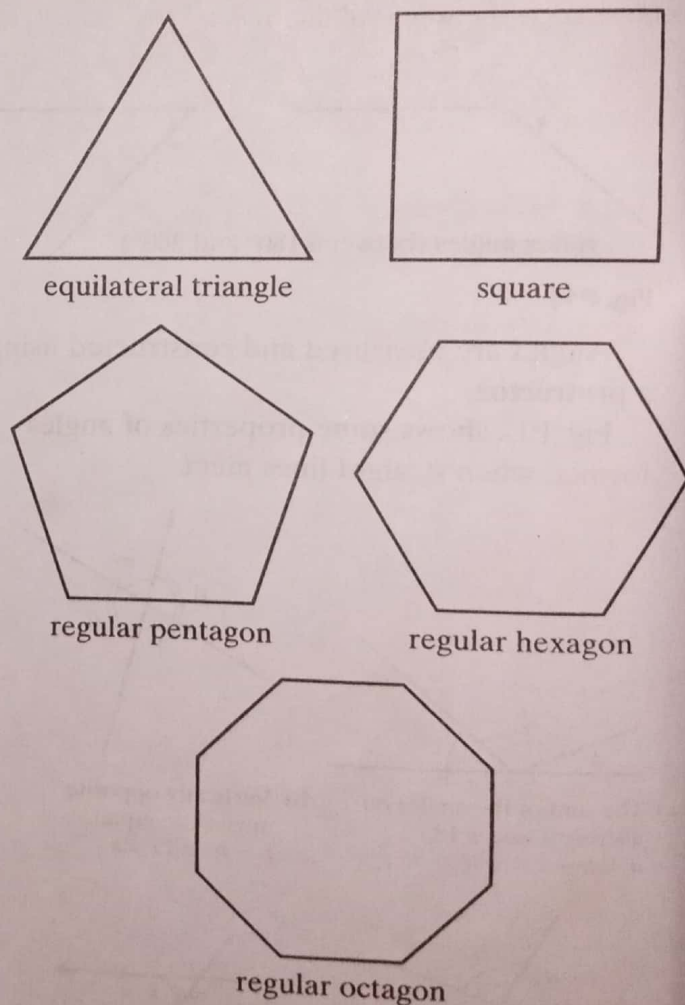


Fig. P19

The **sum of the angles of an n -sided polygon** is $(n - 2) \times 180^\circ$. In particular, the sum of the angles of a triangle is 180° and the sum of the angles of a quadrilateral is 360° .

Any two shapes are **similar** if they are **equiangular** and if their corresponding sides are in the same ratio. Any two triangles are similar if they are equiangular (Fig. P20).

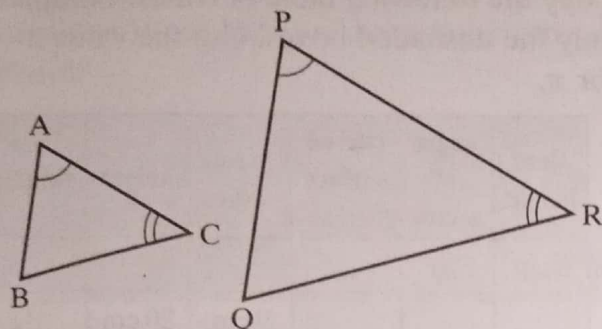


Fig. P20

In Fig. P20, $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR}$

To **solve a triangle** means to calculate the sizes of its sides and angles. Right-angled triangles can be solved using **Pythagoras' rule** and the **tangent, sine** and **cosine** ratios.

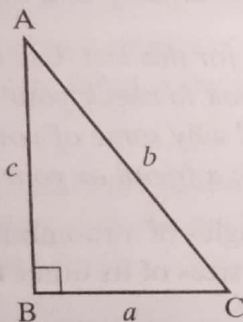


Fig. P21

In Fig. P21, $\triangle ABC$ is right-angled at B; side $\overline{AC}$ is the **hypotenuse**.

Pythagoras' rule: $b^2 = a^2 + c^2$

tangent ratio: $\tan \widehat{C} = \frac{c}{a}$ and $\tan \widehat{A} = \frac{a}{c}$

sine ratio: $\sin \widehat{C} = \frac{c}{b}$ and $\sin \widehat{A} = \frac{a}{b}$

cosine ratio: $\cos \widehat{C} = \frac{a}{b}$ and $\cos \widehat{A} = \frac{c}{b}$

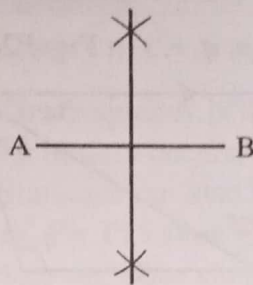
A **pythagorean triple** is a set of three numbers that satisfies Pythagoras' rule. For example {3, 4, 5}, {5, 12, 13}, {8, 15, 17} are pythagorean triples.

Formulae for the **perimeter** and **area of plane shapes** are given in the table on page 233.

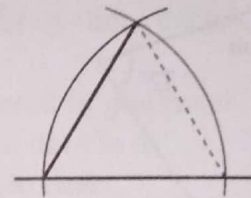
The **SI system of units** is given in the tables on pages 232 and 233.

d The sketches in Fig. P22 show the main features of the common **geometrical constructions**.

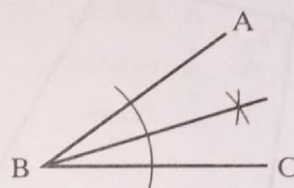
To construct angles of 45° and 30° , bisect angles of 90° and 60° respectively.



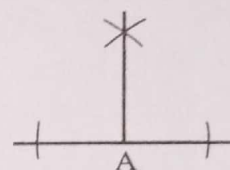
a) perpendicular bisector of AB (ruler and compasses)



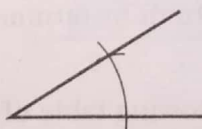
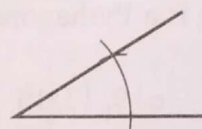
b) equilateral $\triangle$ angle of 60° (ruler and compasses)



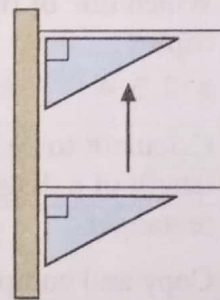
c) bisector of $\widehat{ABC}$ (ruler and compasses)



d) right angle at A (ruler and compasses)



e) to copy an angle (ruler and compasses)



f) parallel lines (ruler and set square)

Fig. P22

Detailed coverage of *geometry and mensuration* is given in the JSS course books as follows:

Book 1, Chapters 6, 8, 11, 13, 14, 16, 20, 21;

Book 2, Chapters 3, 6, 16, 17, 19, 20, 24;

Book 3, Chapters 4, 8, 15, 17, 18, R3, and the SSS course Book 1, Chapters 2, 11, 12, 15, 16, 17.

Review test 5 (Geometry and mensuration)

Allow 40 minutes for this test. Use the answers at the back of the book to check your work. If you do not understand why some of your answers were incorrect, ask a friend or your teacher. Then try Test 6.